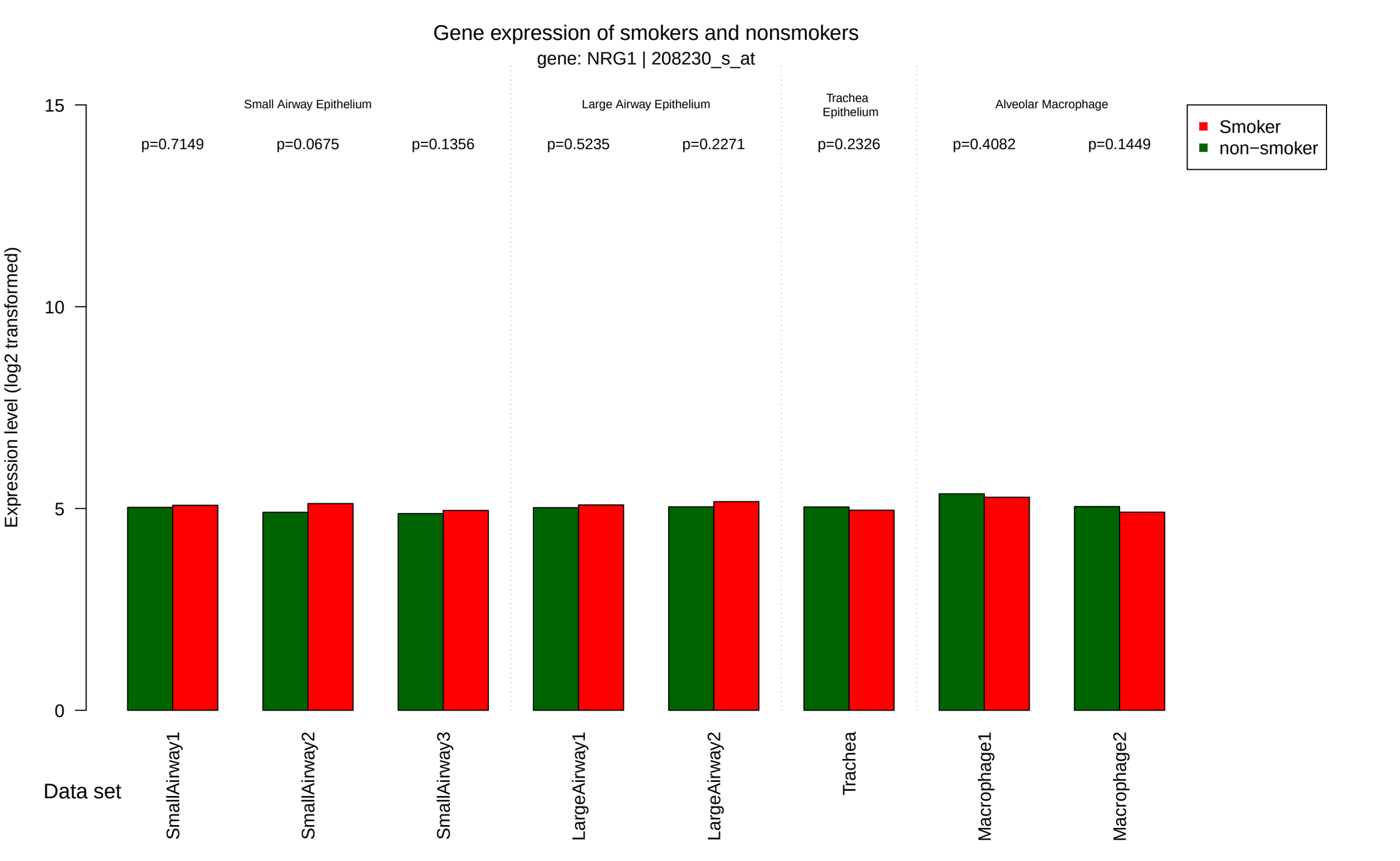
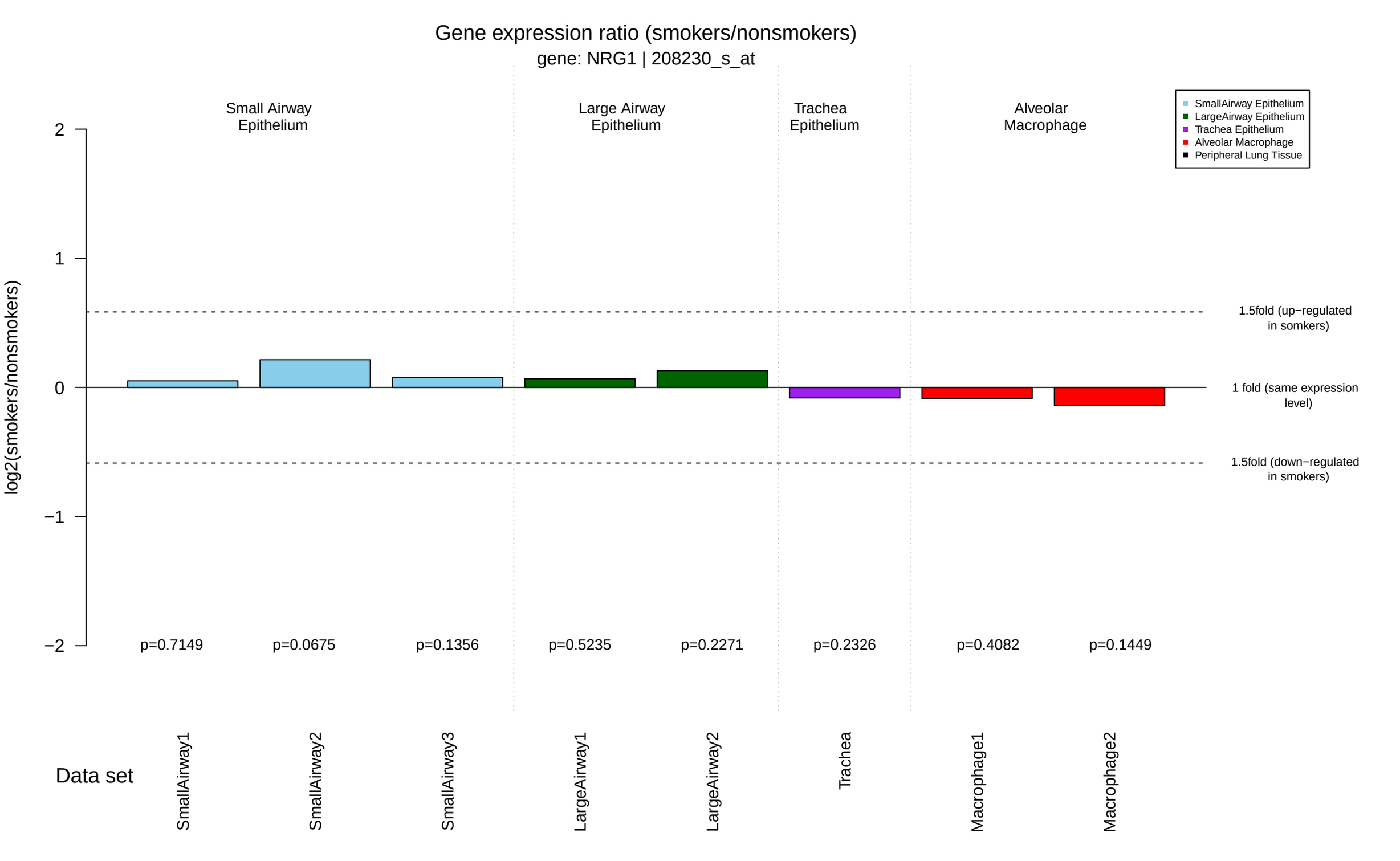
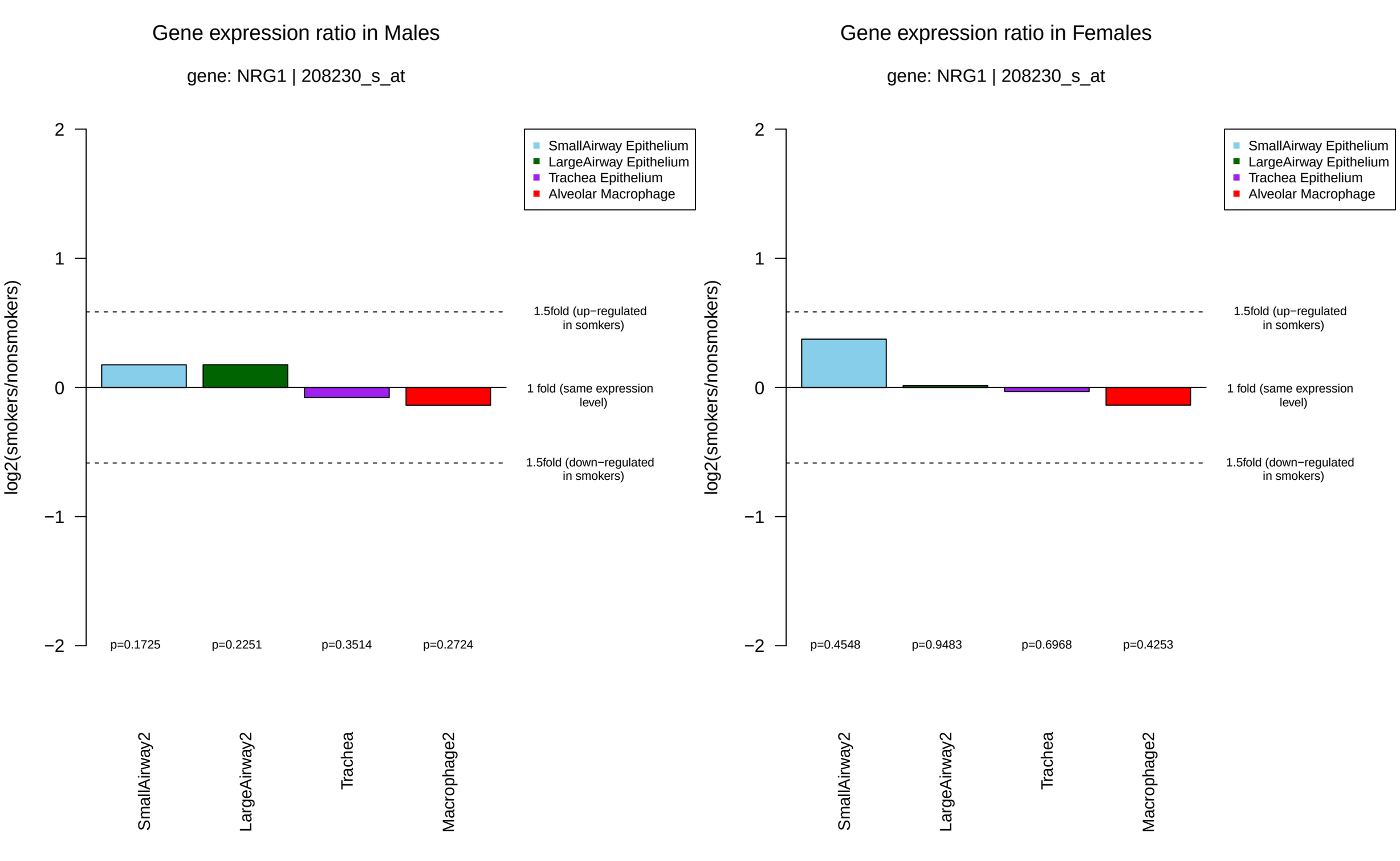
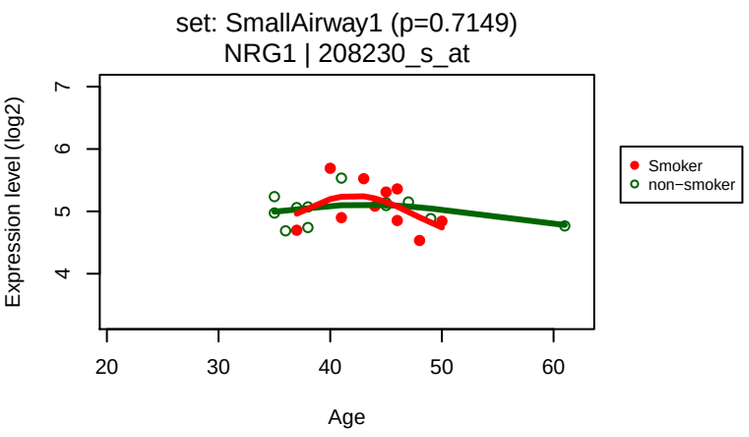
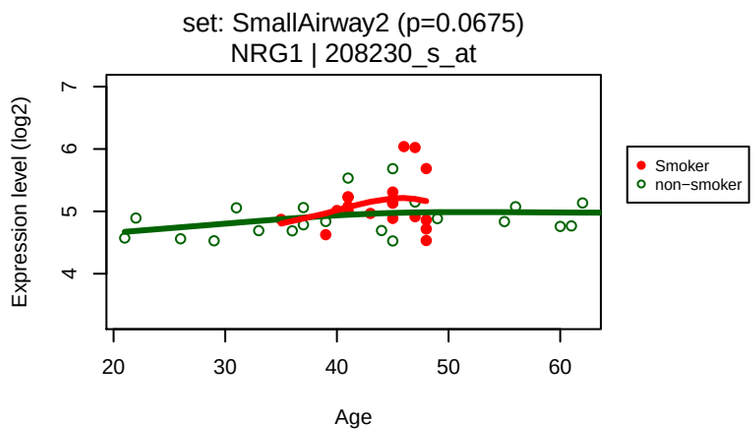
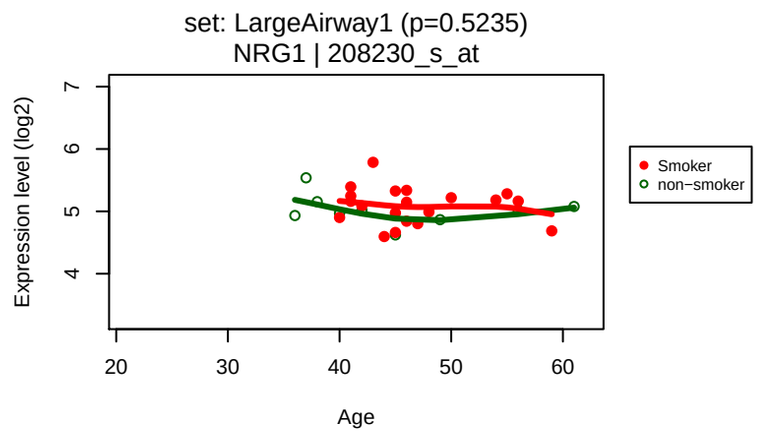

Gene expression of smokers and nonsmokers
gene: NRG1 | 208230_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7149
p=0.0675
p=0.1356
p=0.5235
p=0.2271
p=0.2326
p=0.4082
p=0.1449
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

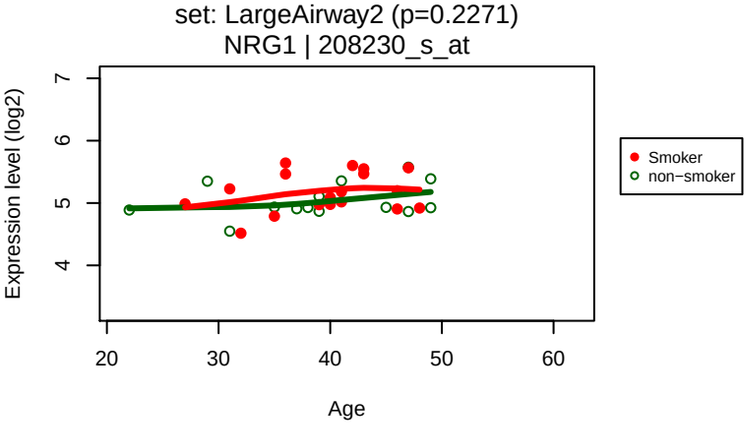

Gene expression ratio (smokers/nonsmokers)
gene: NRG1 | 208230_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7149
p=0.0675
p=0.1356
p=0.5235
p=0.2271
p=0.2326
p=0.4082
p=0.1449
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

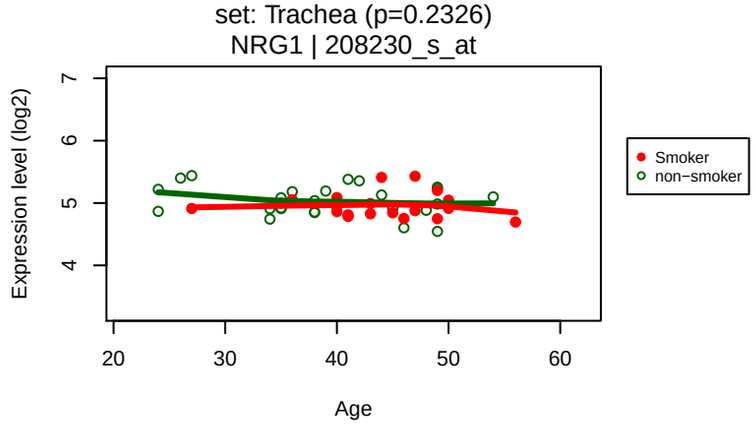

Gene expression ratio in Males
gene: NRG1 | 208230_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1725
p=0.2251
p=0.3514
p=0.2724
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NRG1 | 208230_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4548
p=0.9483
p=0.6968
p=0.4253
SmallAirway2
LargeAirway2
Trachea
Macrophage2

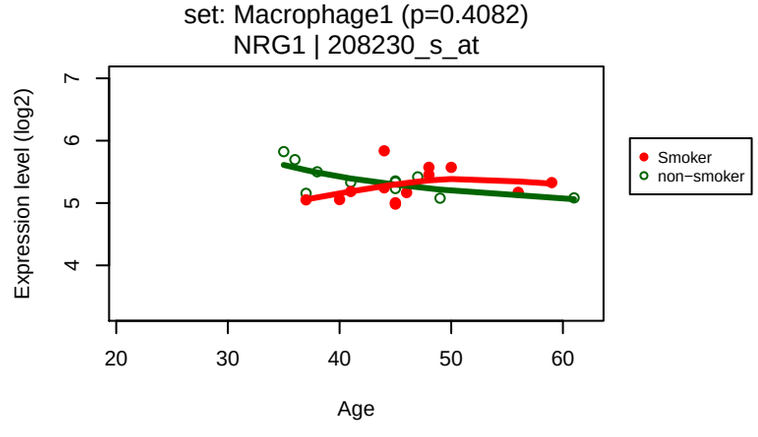

set: SmallAirway1 (p=0.7149)
NRG1 | 208230_s_at
Expression level (log2)
Age
Smoker
non−smoker

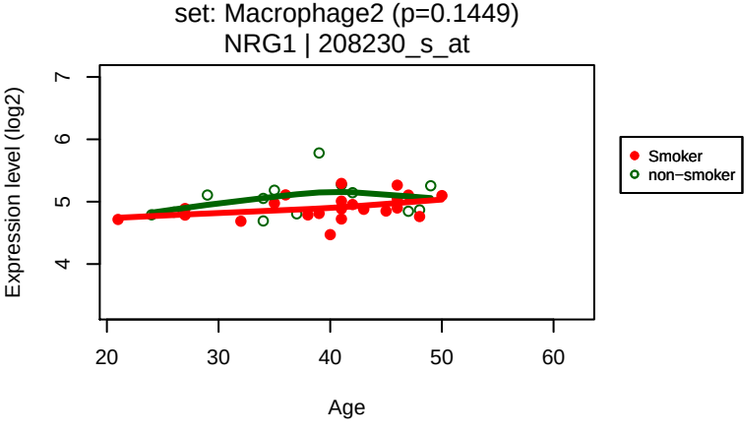

set: SmallAirway2 (p=0.0675)
NRG1 | 208230_s_at
Expression level (log2)
Age
Smoker
non−smoker

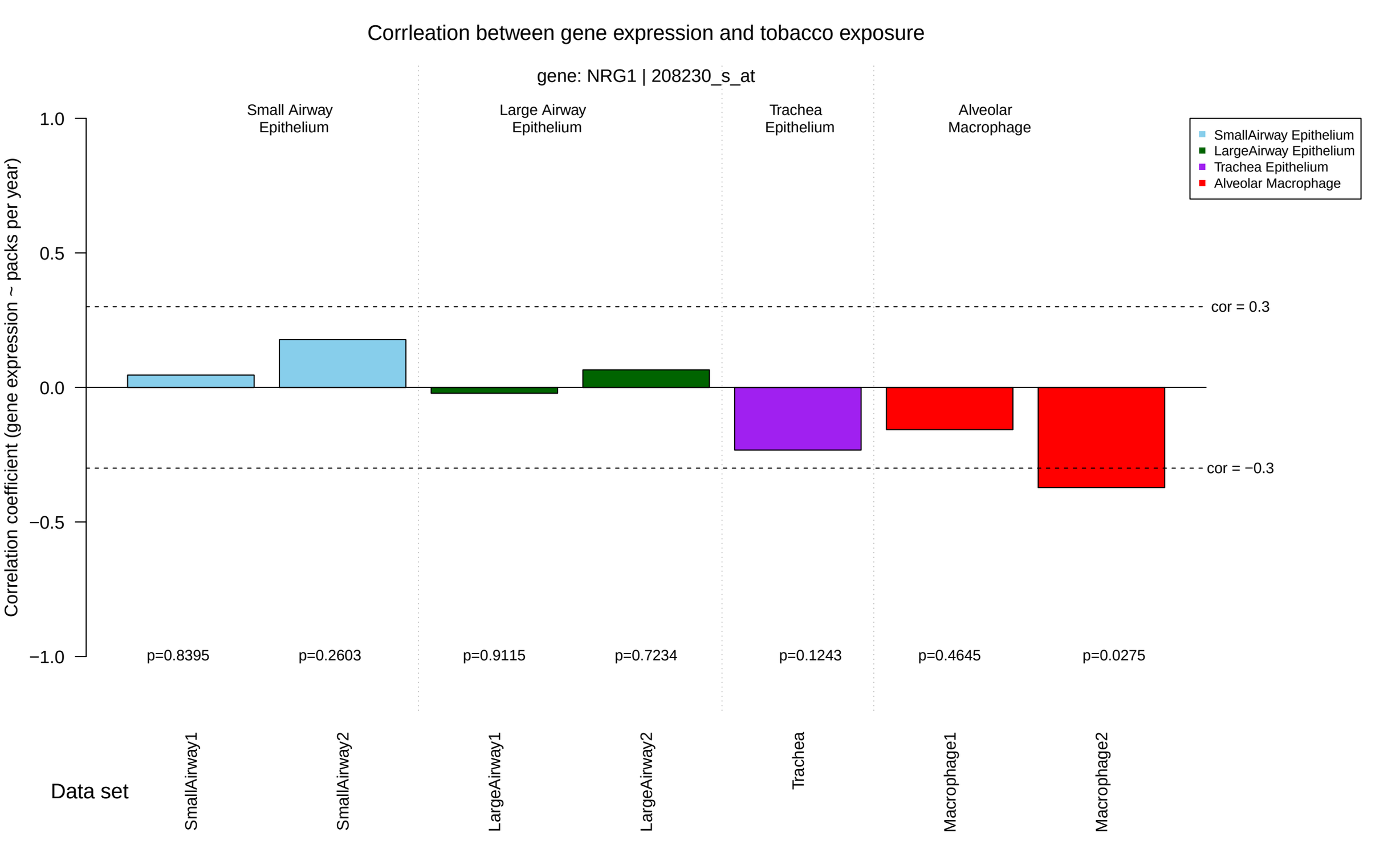

set: LargeAirway1 (p=0.5235)
NRG1 | 208230_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2271)
NRG1 | 208230_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2326)
NRG1 | 208230_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4082)
NRG1 | 208230_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1449)
NRG1 | 208230_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NRG1 | 208230_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8395
p=0.2603
p=0.9115
p=0.7234
p=0.1243
p=0.4645
p=0.0275
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2